

STEAM-AI Diagnostic Report

STEAM_V322_2040, horizon 2040, run STEAM_V322_2040_2040_c42d912d

Generated 2026-09-22T22:30:12+00:00 by STEAM-AI 0.1.0. Ingested 2026-09-22T22:13:54.976077+00:00.

DRAFT: DRAFT - generated by STEAM-AI for modeller review

Run details

Item	Value
Run id	STEAM_V322_2040_2040_c42d912d
Scenario	STEAM_V322_2040
Horizon year	2040
Policy set	REF
STEAM version	3.2.2
Ingested at	2026-09-22T22:13:54.976077+00:00
Report generated at	2026-09-22T22:30:12+00:00
STEAM-AI version	0.1.0

[src: manifest.json]

This report lists what STEAM-AI found when it checked the run, ranked by how much each issue matters. Every number carries a source reference in small grey text. The report is a draft for modeller review and is not a model result.

Executive summary

Run health score: 84.0 out of 100, grade B.

[src: health.json]

Definition: Health = 0.7 x Findings component + 0.3 x Convergence component. Findings component = max(0, 100 - 25 x Critical - 8 x High - 2 x Medium - 0 x Info), counting each distinct problem type (a check and the issue it found) once, at its worst significant severity: a coding error repeated on 8,000 links is one problem. Convergence component = 100 if every assignment period met the relative gap target, scaled down linearly to 0 at ten times the target. When a run has no convergence data, the findings component carries the full weight. Grade: A >= 90, B >= 75, C >= 60, D >= 40, E below 40.

[src: config/health.yaml; health.json]

Health score components

Component	Score	Weight	Detail
Findings	84.0	1.00	100 - 16 penalty over 3 distinct problem type(s) across 23 significant finding(s) (High 22, Info 1); carries full weight because convergence data is unavailable
Convergence	0.0	0.00	no convergence table in this run

[src: health.json]

The checks produced 23 findings: 0 Critical, 22 High, 0 Medium, 1 Info.

[src: findings.json]

Findings by severity

Severity	Findings	Significant
Critical	0	0
High	22	22
Medium	0	0
Info	1	1

[src: findings.json]

Top five findings

Top five findings by severity

Rank	Severity	Location	Finding
1	High	Island of 383 nodes (e.g. node 333155)	A group of 383 nodes is cut off from the rest of the network (main network: 75,451 nodes).
2	High	Island of 792 nodes (e.g. node 334467)	A group of 792 nodes is cut off from the rest of the network (main network: 75,451 nodes); its nearest node (501052) is 15 m from main-network node 501044.
3	High	Island of 367 nodes (e.g. node 333115)	A group of 367 nodes is cut off from the rest of the network (main network: 75,451 nodes).
4	High	Island of 104 nodes (e.g. node 903163)	A group of 104 nodes meets the main network at the same point but under a different node number (node 1201048 sits on node 1201051).
5	High	Island of 705 nodes (e.g. node 52262)	A group of 705 nodes is cut off from the rest of the network (main network: 75,451 nodes).

[src: findings/04703ce29df0; findings/1100bb3ee4a8; findings/15066141cbb2; findings/1a399848e59a; findings/27bb8b8034de]

What to fix first

22 Critical or High findings need action. Fix them in the order below; Critical items block use of the run.

[src: findings.json]

Action list

Severity	Location	Finding	Suggested action	Best measure
High	Island of 383 nodes (e.g. node 333155)	A group of 383 nodes is cut off from the rest of the network (main network: 75,451 nodes).	Add or fix the connecting link(s); trips to zones on the island cannot be assigned.	Repair the network coding (not computable, low confidence)
High	Island of 792 nodes (e.g. node 334467)	A group of 792 nodes is cut off from the rest of the network (main network: 75,451 nodes); its nearest node (501052) is 15 m from main-network node 501044.	Add the connecting link between node 501052 and node 501044, or confirm the gap is intended.	Repair the network coding (not computable, low confidence)
High	Island of 367 nodes (e.g. node 333115)	A group of 367 nodes is cut off from the rest of the network (main network: 75,451 nodes).	Add or fix the connecting link(s); trips to zones on the island cannot be assigned.	Repair the network coding (not computable, low confidence)
High	Island of 104 nodes (e.g. node 903163)	A group of 104 nodes meets the main network at the same point but under a different node number (node 1201048 sits on node 1201051).	Merge node 1201048 into node 1201051 (or renumber the links that use it) and re-check connectivity.	Repair the network coding (not computable, low confidence)
High	Island of 705 nodes (e.g. node 52262)	A group of 705 nodes is cut off from the rest of the network (main network: 75,451 nodes).	Add or fix the connecting link(s); trips to zones on the island cannot be assigned.	Repair the network coding (not computable, low confidence)
High	Island of 152 nodes (e.g. node 190095)	A group of 152 nodes is cut off from the rest of the network (main network: 75,451 nodes); its nearest node (190095) is 20 m from main-network node 190094.	Add the connecting link between node 190095 and node 190094, or confirm the gap is intended.	Repair the network coding (not computable, low confidence)
High	Island of 43 nodes (e.g. node 1207759)	A group of 43 nodes meets the main network at the same point but under a different node number (node 1208861 sits on node 1208865).	Merge node 1208861 into node 1208865 (or renumber the links that use it) and re-check connectivity.	Repair the network coding (not computable, low confidence)
High	Island of 142 nodes (e.g. node 379011)	A group of 142 nodes is cut off from the rest of the network (main network: 75,451 nodes); its nearest node (1077490) is 15 m from main-network node 1077506.	Add the connecting link between node 1077490 and node 1077506, or confirm the gap is intended.	Repair the network coding (not computable, low confidence)
High	Island of 25 nodes (e.g. node 201342)	A group of 25 nodes meets the main network at the same point but under a different node number (node 201343 sits on node 201344).	Merge node 201343 into node 201344 (or renumber the links that use it) and re-check connectivity.	Repair the network coding (not computable, low confidence)
High	2,659 separate network fragments	2,659 separate network fragments show the same problem: cut off from the main network. The 10 most severe are listed individually.	Fix the coding rule that produces this pattern rather than each occurrence; add the connecting link between node 501316 and node 501299, or confirm the gap is intended.	Repair the network coding (not computable, low confidence)

[src: findings/04703ce29df0; findings/1100bb3ee4a8; findings/15066141cbb2; findings/1a399848e59a; findings/27bb8b8034de; findings/2d835fe7ee96; findings/318093607faf; findings/44c8f61c3172; findings/46ae046d6b57; findings/621df77a63a5]

12 further Critical or High findings are listed in the findings section.

[src: findings.json]

Findings by severity

Critical findings (0)

No Critical findings for this run.

High findings (22)

Network connectivity: Island of 383 nodes (e.g. node 333155)

High: A group of 383 nodes is cut off from the rest of the network (main network: 75,451 nodes).

Modeller view: Method: union-find over undirected link graph
Values: issue=disconnected_component, component_size=383, gap_to_main_m=4.796e+04, fragment_node=1002762, nearest_main_node=501056, main_component_size=75451
Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 8503, 8504, 8505, 8506, 8507, 8510, 8513, 8628, 30024, 30027 (+10 more)
SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL
[src: findings/04703ce29df0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8505 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8506 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8513 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30024 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30027 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30032 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30036 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30041 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30042 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30043 col link_id]

Likely cause: A missing link between the island and the main network, or a coding error in end-node ids.

[src: findings/04703ce29df0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8505 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8506 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8513 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30024 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30027 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30032 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30036 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30041 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30042 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30043 col link_id]

Suggested action: Add or fix the connecting link(s); trips to zones on the island cannot be assigned.

[src: findings/04703ce29df0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8505 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8506 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8513 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30024 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30027 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30032 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30036 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30041 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30042 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30043 col link_id]

Evidence

Metric	Value	Threshold
issue	disconnected_component	

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/04703ce29df0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8504 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8505 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8506 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8507 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8513 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30024 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30027 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30029 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30032 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30036 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30041 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30042 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 30043 col link_id]

Network connectivity: Island of 792 nodes (e.g. node 334467)

High: A group of 792 nodes is cut off from the rest of the network (main network: 75,451 nodes); its nearest node (501052) is 15 m from main-network node 501044.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=792, gap_to_main_m=15, fragment_node=501052, nearest_main_node=501044, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 8760, 8761, 8765, 8773, 8774, 8775, 8841, 8935, 13175, 13176 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/1100bb3ee4a8; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8761 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8765 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8773 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8774 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8775 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8935 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13176 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13177 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13178 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13187 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13188 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13189 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13191 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13192 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13193 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13194 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

[src: findings/1100bb3ee4a8; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8761 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8765 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8773 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8774 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8775 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8935 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13176 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13177 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13178 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13187 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13188 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13189 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13190 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13191 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13192 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13193 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13194 col link_id]

Suggested action: Add the connecting link between node 501052 and node 501044, or confirm the gap is intended.

[src: findings/1100bb3ee4a8; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8761 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8765 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8773 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8774 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8775 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8935 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13176 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13177 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13178 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13187 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13188 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13189 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13190 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13191 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13192 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13193 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13194 col link_id]

Evidence

Metric	Value	Threshold
--------	-------	-----------

issue	disconnected_component	
component_size	792	
gap_to_main_m	15	
fragment_node	501,052	
nearest_main_node	501,044	
main_component_size	75,451	
example_nodes	20 items (full list in findings.json): 334,467, 334,469, 334,471	
example_links	20 items (full list in findings.json): 8,760, 8,761, 8,765	

Sources: network 2025 2040.bin.az (v322 AD 20250606 v19. 2040 fields) (links) row 8760 col link id:

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8761 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8765 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8773 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8774 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8775 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8841 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8935 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13175 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13176 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13177 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13178 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13187 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13188 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13189 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13190 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13191 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13192 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13193 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13194 col link_id

src: findings/1100bb3ee4a8; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8761 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2040 fields) (links) row 8765 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8773 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8774 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2040 fields) (links) row 8775 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8841 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8935 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2040 fields) (links) row 13175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13176 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13177 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2040 fields) (links) row 13178 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13187 col link_id;

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13188 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13189 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13190 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13191 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13192 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13193 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13194 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/1100bb3ee4a8; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8761 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8765 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8773 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8774 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8775 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8935 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13176 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13177 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13178 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13187 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13188 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13189 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13190 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13191 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13192 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13193 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13194 col link_id]

Network connectivity: Island of 367 nodes (e.g. node 333115)

High: A group of 367 nodes is cut off from the rest of the network (main network: 75,451 nodes).

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=367, gap_to_main_m=1292, fragment_node=1000655, nearest_main_node=334951, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 8496, 8497, 8498, 8499, 8500, 8551, 8552, 8678, 8681, 28630 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/15066141cbb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8497 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8498 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8499 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8500 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8678 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8681 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28630 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28640 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28649 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28650 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28684 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28685 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28706 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28707 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28718 col link_id]

Likely cause: A missing link between the island and the main network, or a coding error in end-node ids.

[src: findings/15066141cbb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8497 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8498 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8499 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8500 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8678 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8681 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28630 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28640 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28649 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28650 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28684 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28685 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28706 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28707 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28718 col link_id]

Suggested action: Add or fix the connecting link(s); trips to zones on the island cannot be assigned.

[src: findings/15066141cbb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8497 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8498 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8499 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8500 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8678 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8681 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28630 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28640 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28649 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28650 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28684 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28685 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28706 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28707 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28718 col link_id]

Evidence

```
[src: findings/15066141cbb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8497 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8498 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8499 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8500 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8678 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8681 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28630 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28640 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28649 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28650 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28684 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28685 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28706 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28707 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28718 col link_id]
```

```
[src: findings/15066141cbb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8497 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8498 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8499 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8500 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8678 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8681
```

col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28630 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28640 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28649 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28650 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28684 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28685 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28706 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28707 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28718 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/15066141cbb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8497 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8498 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8499 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8500 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8551 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8678 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8681 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28630 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28640 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28649 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28650 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28684 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28685 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28706 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28707 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28718 col link_id]

Network connectivity: Island of 104 nodes (e.g. node 903163)

High: A group of 104 nodes meets the main network at the same point but under a different node number (node 1201048 sits on node 1201051).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=104, gap_to_main_m=0, fragment_node=1201048, nearest_main_node=1201051, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 28100, 116639, 116646, 116647, 116693, 116694, 116716, 116717, 116754, 116755 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/1a399848e59a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116646 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116647 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116716 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116754 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116755 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116756 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116763 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116764 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116786 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116787 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116864 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116865 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116893 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116894 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116940 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/1a399848e59a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116646 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116647 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116716 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116754 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116755 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116756 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116763 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116764 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116786 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116787 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116864 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116865 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116893 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116894 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116940 col link_id]

Suggested action: Merge node 1201048 into node 1201051 (or renumber the links that use it) and re-check connectivity.

[src: findings/1a399848e59a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116646 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116647 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116716 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116754 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116755 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116756 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116763 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116764 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116786 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116787 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116864 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116865 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116893 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116894 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116940 col link_id]

Evidence

Metric	Value	Threshold
issue	node_id_mismatch	
component_size	104	
gap_to_main_m	0	
fragment_node	1,201,048	
nearest_main_node	1,201,051	
main_component_size	75,451	
example_nodes	20 items (full list in findings.json): 903,163, 1,201,044, 1,201,048	
example_links	20 items (full list in findings.json): 28,100, 116,639, 116,646	

[src: findings/1a399848e59a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116646 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116647 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116716 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116754 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116755 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116756 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116763 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116764 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116786 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116787 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116864 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116865 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116893 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116894 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116940 col link_id]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116639 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116646 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116647 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116716 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116754 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116755 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116756 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116763 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116764 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116786 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116787 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116864 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116865 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116893 col link_id;

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116894 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116940 col link_id

[src: findings/1a399848e59a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116639 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116646 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116647 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116693 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116716 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116717 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116754 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116755 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116756 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116763 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116764 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116786 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116787 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116864 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116865 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116893 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116894 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116940 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/1a399848e59a; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116639 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116646 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116647 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116693 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116716 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116717 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116754 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116755 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116756 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116763 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116764 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116786 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116787 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116864 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116865 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 116893 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
116894 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 116940 col link_id]

Network connectivity: Island of 705 nodes (e.g. node 52262)

High: A group of 705 nodes is cut off from the rest of the network (main network: 75,451 nodes).

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=705, gap_to_main_m=78,
fragment_node=501064, nearest_main_node=501024, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 11, 12, 8535, 8536, 8542, 8544, 8545, 8550, 8624, 8686 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/27bb8b8034de; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 12 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8535 col
link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8536 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 8542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8544
col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8545 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 8550 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8624
col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8686 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 8687 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8693
col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8694 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 8697 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
19487 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19488 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 19489 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
19490 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19491 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 19492 col link_id]

Likely cause: A missing link between the island and the main network, or a coding error in end-node ids.

[src: findings/27bb8b8034de; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 12 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8535 col
link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8536 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 8542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8544
col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8545 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 8550 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8624
col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8686 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 8687 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8693

Suggested action: Add or fix the connecting link(s); trips to zones on the island cannot be assigned.

Evidence

Metric	Value	Threshold
issue	disconnected_component	
component_size	705	
gap_to_main_m	78	
fragment_node	501,064	
nearest_main_node	501,024	
main_component_size	75,451	
example_nodes	20 items (full list in findings.json): 52,262, 333,305, 333,306	
example_links	20 items (full list in findings.json): 11, 12, 8,535	

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 12 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8535 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8536 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8542 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8544 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8545 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8550 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8624 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8686 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8687 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8693 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8694 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8697 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19487 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19488 col link_id;

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19489 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19490 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19491 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19492 col link_id

[src: findings/27bb8b8034de; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 12 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8536 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8544 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8545 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8624 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8686 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8687 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8697 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19487 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19488 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19489 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19490 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19491 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19492 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/27bb8b8034de; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 12 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8536 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8544 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8545 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8624 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8686 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8687 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8693 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8694 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8697 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19487 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19488 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19489 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19490 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19491 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19492 col link_id]

Network connectivity: Island of 152 nodes (e.g. node 190095)

High: A group of 152 nodes is cut off from the rest of the network (main network: 75,451 nodes); its nearest node (190095) is 20 m from main-network node 190094.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=152, gap_to_main_m=20.3, fragment_node=190095, nearest_main_node=190094, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 3946, 10272, 10273, 10274, 10275, 102976, 102978, 102979, 102980, 102981 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/2d835fe7ee96; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 3946 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10272 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10273 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10274 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10275 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102976 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102978 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102984 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102985 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102992 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102993 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103039 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103042 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

[src: findings/2d835fe7ee96; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 3946 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10272 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row

Suggested action: Add the connecting link between node 190095 and node 190094, or confirm the gap is intended.

Evidence

Evidence		
Metric	Value	Threshold
issue	disconnected_component	
component_size	152	
gap_to_main_m	20.3	
fragment_node	190,095	
nearest_main_node	190,094	
main_component_size	75,451	
example_nodes	20 items (full list in findings.json): 190,095, 190,096, 375,507	
example_links	20 items (full list in findings.json): 3,946, 10,272, 10,273	

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 3946 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10272 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10273 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10274 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10275 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102976 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102978 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102979 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102980 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102981 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102982 col link_id;

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102984 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102985 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102992 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102993 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103033 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103034 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103039 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103040 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103042 col link_id

[src: findings/2d835fe7ee96; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 3946 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10272 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10273 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10274 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10275 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102976 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102978 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102984 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102985 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102992 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102993 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103039 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103042 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/2d835fe7ee96; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 3946 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10272 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10273 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10274 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10275 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102976 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102978 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102984 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102985 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102992 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 102993 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103033 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103034 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103039 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103040 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 103042 col link_id]

Network connectivity: Island of 43 nodes (e.g. node 1207759)

High: A group of 43 nodes meets the main network at the same point but under a different node number (node 1208861 sits on node 1208865).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=43, gap_to_main_m=0, fragment_node=1208861, nearest_main_node=1208865, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 120355, 120362, 120368, 120403, 120404, 120406, 120407, 120411, 120412, 120416 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/318093607faf; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120355 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120362 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120368 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120403 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120404 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120406 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120407 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120411 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120412 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120416 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120417 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120421 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120425 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120426 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120427 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120428 col link_id; network_2025_2040.bin.gz

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120406 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120407 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120411 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120412 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120416 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120417 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120420 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120421 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120425 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120426 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120427 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120428 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120429 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120430 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120438 col link_id

[src: findings/318093607faf; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120355 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120362 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120368 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120403 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120404 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120406 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120407 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120411 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120412 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120416 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120417 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120421 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120425 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120426 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120427 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120428 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120429 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120430 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120438 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/318093607faf; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120355 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120362 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120368 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120403 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120404 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120406 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120407 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120411 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120412 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120416 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120417 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120421 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120425 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120426 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120427 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120428 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 120429 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
120430 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120438 col link_id]

Network connectivity: Island of 142 nodes (e.g. node 379011)

High: A group of 142 nodes is cut off from the rest of the network (main network: 75,451 nodes); its nearest node (1077490) is 15 m from main-network node 1077506.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=142, gap_to_main_m=15, fragment_node=1077490, nearest_main_node=1077506, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 10613, 10614, 10615, 10616, 10617, 10618, 10619, 10620, 10621, 10622 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/44c8f61c3172; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10613 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10614 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
2040 fields) (links) row 10615 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10616 col link_id;

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

Suggested action: Add the connecting link between node 1077490 and node 1077506, or confirm the gap is intended.

Evidence

```
[src: findings/44c8f61c3172; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10613 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10614 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10615 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10616 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10617 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10618 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10619 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10620 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10621 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10622 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10623 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10624 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10625 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10626 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10627 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10629 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10630 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10631 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10632 col link_id]
```

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10613 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10614 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10615 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10616 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10617 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10618 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10619 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10620 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10621 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10622 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10623 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10624 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10625 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10626 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10627 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10628 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10629 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10630 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10631 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10632 col link_id

[src: findings/44c8f61c3172; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10613 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10614 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10615 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10616 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10617 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10618 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10619 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10620 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10621 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10622 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10623 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10624 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10625 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10626 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10627 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10628 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10629 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10630 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10631 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10632 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/44c8f61c3172; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10613 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10614 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10615 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10616 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10617 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10618 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10619 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10620 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10621 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10622 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10623 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10624 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10625 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10626 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10627 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10628 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10629 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10630 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10631 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10632 col link_id]

Network connectivity: Island of 25 nodes (e.g. node 201342)

High: A group of 25 nodes meets the main network at the same point but under a different node number (node 201343 sits on node 201344).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=25, gap_to_main_m=0, fragment_node=201343, nearest_main_node=201344, main_component_size=75451

High: 2,659 separate network fragments show the same problem: cut off from the main network. The 10 most severe are listed individually.

Modeller view: Method: 2,659 findings of issue 'disconnected_component' grouped; the 10 most severe kept as individual findings

Values: issue=disconnected_component, n_occurrences=2659, n_listed=10, by_severity={'High': 2659}, location_type=node, nodes_in_fragments=25478

Thresholds: group_issue_above=25, group_keep_worst=10

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 7, 8760, 11, 8503, 8496

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/621df77a63a5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

[src: findings/621df77a63a5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id]

Suggested action: Fix the coding rule that produces this pattern rather than each occurrence; add the connecting link between node 501316 and node 501299, or confirm the gap is intended.

[src: findings/621df77a63a5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id]

Evidence

Metric	Value	Threshold
issue	disconnected_component	
n_occurrences	2,659	
n_listed	10	
by_severity	High=2,659	
location_type	node	
nodes_in_fragments	25,478	
example_ids	50 items (full list in findings.json): 52204, 334467, 52262	
summarised_ids	500 items (full list in findings.json): 202775, 1016876, 200891	
group_issue_above		25
group_keep_worst		10

[src: findings/621df77a63a5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8503 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id

[src: findings/621df77a63a5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/621df77a63a5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8760 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 11 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8503 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8496 col link_id]

Network connectivity: Island of 63 nodes (e.g. node 202703)

High: A group of 63 nodes meets the main network at the same point but under a different node number (node 1243673 sits on node 1243671).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=63, gap_to_main_m=0, fragment_node=1243673, nearest_main_node=1243671, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 7556, 7559, 140834, 140835, 140845, 140846, 140891, 140892, 140922, 140923 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/70cc58836fb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7559 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140892 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140923 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140964 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140999 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141000 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141001 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141002 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141003 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141004 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/70cc58836fb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7559 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140892 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140923 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140964 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140999 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141000 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141001 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141002 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141003 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141004 col link_id]

Suggested action: Merge node 1243673 into node 1243671 (or renumber the links that use it) and re-check connectivity.

[src: findings/70cc58836fb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7559 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140892 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140923 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140964 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140999 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141000 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141001 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141002 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141003 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141004 col link_id]

Evidence

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/70cc58836fb2; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140834 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140835 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140846 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140891 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140892 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140923 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140964 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 140999 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141000 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141001 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141002 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141003 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 141004 col link_id]

Network connectivity: Island of 181 nodes (e.g. node 700645)

High: A group of 181 nodes is cut off from the rest of the network (main network: 75,451 nodes).

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=181, gap_to_main_m=124.5,

fragment_node=813183, nearest_main_node=813181, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 19696, 26939, 26952, 26956, 26966, 26967, 26972, 26973, 26974, 26975 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/a17b1d687800; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19696 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26939 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26952 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26956 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26966 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26967 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26972 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26973 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26974 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26975 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26976 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26977 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26978 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26983 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26984 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26998 col link_id]

Likely cause: A missing link between the island and the main network, or a coding error in end-node ids.

[src: findings/a17b1d687800; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19696 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26939 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26952 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26956 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26966 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26967 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26972 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26973 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26974 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26975 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26976 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26977 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26978 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26983 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26984 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26998 col link_id]

Suggested action: Add or fix the connecting link(s); trips to zones on the island cannot be assigned.

[src: findings/a17b1d687800; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19696 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26939 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26952 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26956 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26966 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26967 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26972 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26973 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26974 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26975 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26976 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26977 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26978 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26983 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26984 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26998 col link_id]

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26998 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/a17b1d687800; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 19696 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26939 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26952 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26956 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26966 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26967 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26972 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26973 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26974 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26975 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26976 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26977 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26978 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26979 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26980 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26981 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26982 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26983 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26984 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 26998 col link_id]

Network connectivity: Island of 38 nodes (e.g. node 1099812)

High: A group of 38 nodes meets the main network at the same point but under a different node number (node 1100999 sits on node 1101006).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=38, gap_to_main_m=0, fragment_node=1100999, nearest_main_node=1101006, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 72924, 73028, 73031, 73057, 73064, 73066, 73067, 73074, 73075, 73081 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/a81f145e2e3c; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 72924 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73031 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73057 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73067 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73074 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73075 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73081 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73106 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73155 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73173 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73229 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/a81f145e2e3c; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 72924 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73031 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73057 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73067 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73074 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73075 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73081 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73106 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73155 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73173 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73229 col link_id]

Suggested action: Merge node 1100999 into node 1101006 (or renumber the links that use it) and re-check connectivity.

[src: findings/a81f145e2e3c; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 72924 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73031 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73057 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73067 col link_id;

Evidence

```
[src: findings/a81f145e2e3c; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 72924 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73031 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73057 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73067 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73074 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73075 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73081 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73106 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73155 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73173 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73229 col link_id]
```

```
[src: findings/a81f145e2e3c; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 72924 col link_id;  
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,  
2040 fields) (links) row 73031 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73057 col link_id;  
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,  
2040 fields) (links) row 73066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73067 col link_id;  
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73074 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19,
```


2040 fields) (links) row 73075 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73081 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73106 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73155 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73173 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73229 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/a81f145e2e3c; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 72924 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73028 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73031 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73057 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73064 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73067 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73074 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73075 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73081 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73106 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73155 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73173 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 73229 col link_id]

Network connectivity: Island of 25 nodes (e.g. node 901407)

High: A group of 25 nodes meets the main network at the same point but under a different node number (node 1196318 sits on node 1196320).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=25, gap_to_main_m=0, fragment_node=1196318, nearest_main_node=1196320, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 27805, 113419, 113420, 113483, 113484, 113542, 113543, 113555, 113556, 113568 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/b6462f4acefa; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 27805 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113483 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113484 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113543 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113555 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113568 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113569 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113578 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113579 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113599 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113600 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113603 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113604 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/b6462f4acefa; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 27805 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113483 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113484 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113543 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113555 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113568 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113569 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113578 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113579 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113599 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113600 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113603 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113604 col link_id]

Suggested action: Merge node 1196318 into node 1196320 (or renumber the links that use it) and re-check connectivity.

[src: findings/b6462f4acefa; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 27805 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113483 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113484 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113543 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113555 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113568 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113569 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113578 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113579 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113599 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113600 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113603 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113604 col link_id]

Evidence

Metric	Value	Threshold
issue	node_id_mismatch	
component_size	25	
gap_to_main_m	0	
fragment_node	1,196,318	
nearest_main_node	1,196,320	
main_component_size	75,451	
example_nodes	20 items (full list in findings.json): 901,407, 901,422, 1,195,720	
example_links	20 items (full list in findings.json): 27,805, 113,419, 113,420	

[src: findings/b6462f4acefa; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 27805 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113483 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113484 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113543 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113555 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113568 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113569 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113578 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113579 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113599 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113600 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113603 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113604 col link_id]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 27805 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113483 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113484 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113543 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113555 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113568 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113569 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113578 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113579 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113599 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113600 col link_id;

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113603 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113604 col link_id

[src: findings/b6462f4acefa; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 27805 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113483 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113484 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113543 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113555 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113568 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113569 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113578 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113579 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113599 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113600 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113603 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113604 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/b6462f4acefa; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 27805 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113420 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113483 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113484 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113542 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113543 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113555 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113568 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113569 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113578 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113579 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113599 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113600 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113603 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 113604 col link_id]

Network connectivity: Island of 2853 nodes (e.g. node 52204)

High: A group of 2,853 nodes is cut off from the rest of the network (main network: 75,451 nodes); its nearest node (501316) is 11 m from main-network node 501299.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=2853, gap_to_main_m=11.1,

fragment_node=501316, nearest_main_node=501299, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 7, 8, 9, 10, 6461, 6462, 6463, 8217, 8218, 8219 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/b782b172f2e5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 9 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6463 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8217 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8218 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8219 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8223 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8224 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8229 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

[src: findings/b782b172f2e5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 9 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6463 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8217 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8218 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8219 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8223 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8224 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8229 col link_id]

Suggested action: Add the connecting link between node 501316 and node 501299, or confirm the gap is intended.

Evidence

```

[src: findings/b782b172f2e5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 9 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6463 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8217 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8218 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8219 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8223 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8224 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8229 col link_id]

```

STEAM-AI Diagnostic Report | run STEAM V322 2040 2040 c42d912d | generated 2026-09-22T22:30:12+00:00 by STEAM-AI v0.3.3

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8223 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8224 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8225 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8226 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8227 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8228 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8229 col link_id

[src: findings/b782b172f2e5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 9 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6463 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8217 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8218 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8219 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8223 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8224 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8229 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/b782b172f2e5; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 9 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 10 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6461 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6462 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6463 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8217 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8218 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8219 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8220 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8223 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8224 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8225 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8228 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 8229 col link_id]

Network connectivity: Island of 24 nodes (e.g. node 1138358)

High: A group of 24 nodes meets the main network at the same point but under a different node number (node 1138542 sits on node 1138557).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=24, gap_to_main_m=0, fragment_node=1138542, nearest_main_node=1138557, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 89362, 89368, 89369, 89382, 89383, 89384, 89413, 89414, 89418, 89419 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/bd2e8b38afa0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89362 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89368 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89369 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89382 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89383 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89384 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89413 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89414 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89418 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89419 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89423 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89424 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89425 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89435 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89436 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89437 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89438 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89439 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89440 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89454 col link_id]

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89418 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89419 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89423 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89424 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89425 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89435 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89436 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89437 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89438 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89439 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89440 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89454 col link_id

[src: findings/bd2e8b38afa0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89362 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89368 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89369 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89382 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89383 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89384 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89413 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89414 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89418 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89419 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89423 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89424 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89425 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89435 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89436 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89437 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89438 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89439 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89440 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89454 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/bd2e8b38afa0; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89362 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89368 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89369 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89382 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89383 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89384 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89413 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89414 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89418 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89419 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89423 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89424 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89425 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89435 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89436 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89437 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89438 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89439 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89440 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 89454 col link_id]

Network connectivity: Island of 41 nodes (e.g. node 1217101)

High: A group of 41 nodes meets the main network at the same point but under a different node number (node 1217101 sits on node 1217085).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=41, gap_to_main_m=0, fragment_node=1217101, nearest_main_node=1217085, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 126342, 126369, 126395, 126396, 126408, 126409, 126415, 126448, 126449, 126535 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/ce7203cf1732; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126342 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126369 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126395 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126396 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126408 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126409 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126415 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126448 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126449 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126536 col link_id; network_2025_2040.bin.gz

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126395 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126396 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126408 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126409 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126415 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126448 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126449 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126535 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126536 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126540 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126541 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126551 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126552 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126571 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126572 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126584 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126585 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126587 col link_id

[src: findings/ce7203cf1732; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126342 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126369 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126395 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126396 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126408 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126409 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126415 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126448 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126449 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126536 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126540 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126541 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126551 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126571 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126572 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126584 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126585 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126587 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/ce7203cf1732; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126342 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126369 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126395 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126396 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126408 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126409 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126415 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126448 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126449 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126535 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126536 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126540 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126541 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126551 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126552 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126571 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126572 col link_id; network_2025_2040.bin.gz
(v322_AD_20250606_v19, 2040 fields) (links) row 126584 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row
126585 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 126587 col link_id]

Network connectivity: Island of 50 nodes (e.g. node 1191908)

High: A group of 50 nodes meets the main network at the same point but under a different node number (node 1191908 sits on node 1191904).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=50, gap_to_main_m=0, fragment_node=1191908, nearest_main_node=1191904, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 111246, 111253, 111254, 111263, 111264, 111291, 111292, 111293, 111301, 111302 (+10 more)

High: A group of 111 nodes meets the main network at the same point but under a different node number (node 1151491 sits on node 1151474).

Modeller view: Method: union-find over undirected link graph

Values: issue=node_id_mismatch, component_size=111, gap_to_main_m=0, fragment_node=1151491, nearest_main_node=1151474, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 95401, 95510, 95717, 95722, 95899, 95900, 95915, 95922, 96006, 96007 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/ec678e63546f; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95722 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95899 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95900 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95915 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96006 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96007 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96062 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96234 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96239 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/ec678e63546f; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95722 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95899 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95900 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95915 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96006 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96007 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96062 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96234 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96239 col link_id]

Suggested action: Merge node 1151491 into node 1151474 (or renumber the links that use it) and re-check connectivity.

[src: findings/ec678e63546f; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95510 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95717 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95722 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95899 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95900 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95915 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95922 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96006 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96007 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96062 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96234 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96239 col link_id]

Evidence

Metric	Value	Threshold
issue	node_id_mismatch	
component_size	111	
gap_to_main_m	0	
fragment_node	1,151,491	
nearest_main_node	1,151,474	
main_component_size	75,451	
example_nodes	20 items (full list in findings.json): 725,210, 1,151,491, 1,151,616	

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95401 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95510 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95717 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95722 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95899 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95900 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95915 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95922 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96006 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96007 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96062 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96066 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96172 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96175 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96221 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96222 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96226 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96227 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96234 col link_id;
network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96239 col link_id

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

STEAM-AI Diagnostic Report | run STEAM V322 2040 2040 c42d912d | generated 2026-09-22T22:30:12+00:00 by STEAM-AI Page 42

2040 fields) (links) row 96066 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96172 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96175 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96221 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96222 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96226 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96227 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96234 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 96239 col link_id]

Network connectivity: 94 network fragments

High: 94 network fragments show the same problem: touching the main network under a different node number. The 10 most severe are listed individually.

Modeller view: Method: 94 findings of issue 'node_id_mismatch' grouped; the 10 most severe kept as individual findings

Values: issue=node_id_mismatch, n_occurrences=94, n_listed=10, by_severity={'High': 94}, location_type=node, nodes_in_fragments=1036

Thresholds: group_issue_above=25, group_keep_worst=10

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 95401, 28100, 7556, 111246, 120355

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/f05837f34e4d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 111246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120355 col link_id]

Likely cause: Two node numbers were coded at one location, so the links never join.

[src: findings/f05837f34e4d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 111246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120355 col link_id]

Suggested action: Fix the coding rule that produces this pattern rather than each occurrence; merge node 1151491 into node 1151474 (or renumber the links that use it) and re-check connectivity.

[src: findings/f05837f34e4d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 111246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120355 col link_id]

Evidence

Metric	Value	Threshold
issue	node_id_mismatch	
n_occurrences	94	
n_listed	10	
by_severity	High=94	
location_type	node	
nodes_in_fragments	1,036	
example_ids	50 items (full list in findings.json): 725210, 903163, 202703	
summarised_ids	84 items (full list in findings.json): 193907, 1075499, 1133108	
group_issue_above		25
group_keep_worst		10

[src: findings/f05837f34e4d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 111246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120355 col link_id]

Sources: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 111246 col link_id;

network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120355 col link_id]

[src: findings/f05837f34e4d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 111246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120355 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/f05837f34e4d; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 95401 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 28100 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7556 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 111246 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 120355 col link_id]

Network connectivity: Island of 234 nodes (e.g. node 104248)

High: A group of 234 nodes is cut off from the rest of the network (main network: 75,451 nodes); its nearest node (1082259) is 15 m from main-network node 1082342.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=234, gap_to_main_m=15, fragment_node=1082259, nearest_main_node=1082342, main_component_size=75451
Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 1624, 1628, 1629, 4050, 4297, 4298, 63840, 63841, 63844, 63845 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/f7380be69b83; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1624 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1629 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4050 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4297 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4298 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63840 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63844 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63848 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63849 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63850 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63853 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63854 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63859 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63860 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63884 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63922 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

[src: findings/f7380be69b83; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1624 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1629 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4050 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4297 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4298 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63840 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63844 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63848 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63849 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63850 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63853 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63854 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63859 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63860 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63884 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63922 col link_id]

Suggested action: Add the connecting link between node 1082259 and node 1082342, or confirm the gap is intended.

[src: findings/f7380be69b83; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1624 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1629 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4050 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4297 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4298 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63840 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63844 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63848 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63849 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63850 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63853 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63854 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63859 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63860 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63884 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63922 col link_id]

Evidence

```
[src: findings/f7380be69b83; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1624 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1629 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4050 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4297 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4298 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63840 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63844 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63848 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63849 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63850 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63853 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63854 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63859 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63860 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63884 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63922 col link_id]
```

```
[src: findings/f7380be69b83; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1624 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1629 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4050 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4297 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4298 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63840 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63844 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63848 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63849 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63850 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63854 col link_id; network_2025_2040.bin.gz
```


(v322_AD_20250606_v19, 2040 fields) (links) row 63859 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63860 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63884 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63922 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/f7380be69b83; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1624 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1628 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 1629 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4050 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4297 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 4298 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63840 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63841 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63844 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63845 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63848 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63849 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63850 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63851 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63853 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63854 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63859 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63860 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63884 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 63922 col link_id]

Network connectivity: Island of 262 nodes (e.g. node 200743)

High: A group of 262 nodes is cut off from the rest of the network (main network: 75,451 nodes); its nearest node (403743) is 22 m from main-network node 403744.

Modeller view: Method: union-find over undirected link graph

Values: issue=disconnected_component, component_size=262, gap_to_main_m=22.1,

fragment_node=403743, nearest_main_node=403744, main_component_size=75451

Files: network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) rows 6905, 7575, 7608, 13570, 13576, 13577, 143238, 143239, 143270, 143273 (+10 more)

SQL: SELECT a_node, b_node FROM links WHERE a_node IS NOT NULL AND b_node IS NOT NULL

[src: findings/fbd75945f7f8; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6905 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7608 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13570 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143238 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143239 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143273 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143280 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143281 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143284 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143294 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143303 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143304 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143345 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143349 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143350 col link_id]

Likely cause: A junction or connecting link is missing at the gap, so the fragment stops short of the main network.

[src: findings/fbd75945f7f8; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6905 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7608 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13570 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143238 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143239 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143273 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143280 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143281 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143284 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143294 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143303 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143304 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143345 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143349 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143350 col link_id]

Suggested action: Add the connecting link between node 403743 and node 403744, or confirm the gap is intended.

Evidence

```
[src: findings/fbd75945f7f8; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6905 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7608 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13570 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143238 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143239 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143270 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143273 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143280 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143281 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143284 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143294 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143303 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143304 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143345 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143349 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143350 col link_id]
```

[src: findings/fbd75945f7f8; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6905 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7608

col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13570 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143238 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143270 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143273 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143280 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143281 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143284 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143294 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143303 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143304 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143345 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143349 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143350 col link_id]

Proposed measures

Measure	Description	Estimated effect	Method	Confidence
Repair the network coding	Fix the attribute or add the missing link/one-way direction and re-assign.	Not estimated	not computable	low

[src: findings/fbd75945f7f8; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 6905 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7575 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 7608 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13570 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13576 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 13577 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143238 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143270 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143273 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143280 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143281 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143284 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143293 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143294 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143303 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143304 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143345 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143349 col link_id; network_2025_2040.bin.gz (v322_AD_20250606_v19, 2040 fields) (links) row 143350 col link_id]

Medium findings (0)

No Medium findings for this run.

Info findings (1)

Demand matrix sanity: All demand matrices

Info: Trips leaving and entering differ more than 5 to one in 1,446 of 3,196 zones (45%), the pattern of a production-attraction matrix. Per-zone imbalance is not flagged; confirm the matrix format.

Modeller view: Method: share of zones whose row and column totals differ beyond the ratio
Values: issue=pa_format_suspected, zones_imbalanced=1446, zones_checked=3196, share_imbalanced=0.4524
Thresholds: row_col_ratio_max=5, pa_format_share=0.25, min_zone_total_for_ratio=50
Files: od.csv
SQL: WITH rows_ AS (SELECT origin AS zone_id, SUM(trips) AS productions FROM od WHERE matrix_kind = 'DEMAND' GROUP BY 1), cols AS (SELECT destination AS zone_id, SUM(trips) AS attractions FROM od WHERE matrix_kind = 'DEMAND' GROUP BY 1) SELECT zone_id, COALESCE(productions, 0) AS productions, COALESCE(attractions, 0) AS attractions, GREATEST(COALESCE(productions, 0), COALESCE(attractions, 0)) / NULLIF(LEAST(COALESCE(productions, 0), COALESCE(attractions, 0)), 0) AS ratio FROM rows_ FULL OUTER JOIN cols USING (zone_id) WHERE GREATEST(COALESCE(productions, 0), COALESCE(attractions, 0)) >= 50.0 AND (LEAST(COALESCE(productions, 0), COALESCE(attractions, 0)) = 0 OR GREATEST(COALESCE(productions, 0), COALESCE(attractions, 0)) / LEAST(COALESCE(productions, 0), COALESCE(attractions, 0)) > 5.0) ORDER BY ratio DESC NULLS FIRST, zone_id
[src: findings/d538c3a7f464; od.csv (od) col trips]

Likely cause: The matrix is in production-attraction form (trips by home and activity end) rather than origin-destination form.

[src: findings/d538c3a7f464; od.csv (od) col trips]

Suggested action: Confirm the format with the export. If it is meant to be an origin-destination matrix, check the PA-to-OD conversion.

[src: findings/d538c3a7f464; od.csv (od) col trips]

Evidence

Metric	Value	Threshold
issue	pa_format_suspected	
zones_imbalanced	1,446	
zones_checked	3,196	
share_imbalanced	45.2%	
worst_zones	20 items (full list in findings.json): 282, 379, 801	
row_col_ratio_max		5
pa_format_share		25.0%
min_zone_total_for_ratio		50.00

[src: findings/d538c3a7f464; od.csv (od) col trips]

Sources: od.csv (od) col trips

[src: findings/d538c3a7f464; od.csv (od) col trips]

Convergence and noise

Check status: skipped; findings: 0; message: required table(s) missing: convergence.

[src: check_results.json]

The convergence check raised no findings.

[src: findings.json]

No convergence table was exported with this run, so iteration history is not available.

No noise band has been computed for this run (derived/noise_band.parquet is missing), so significance tests use the default rule.

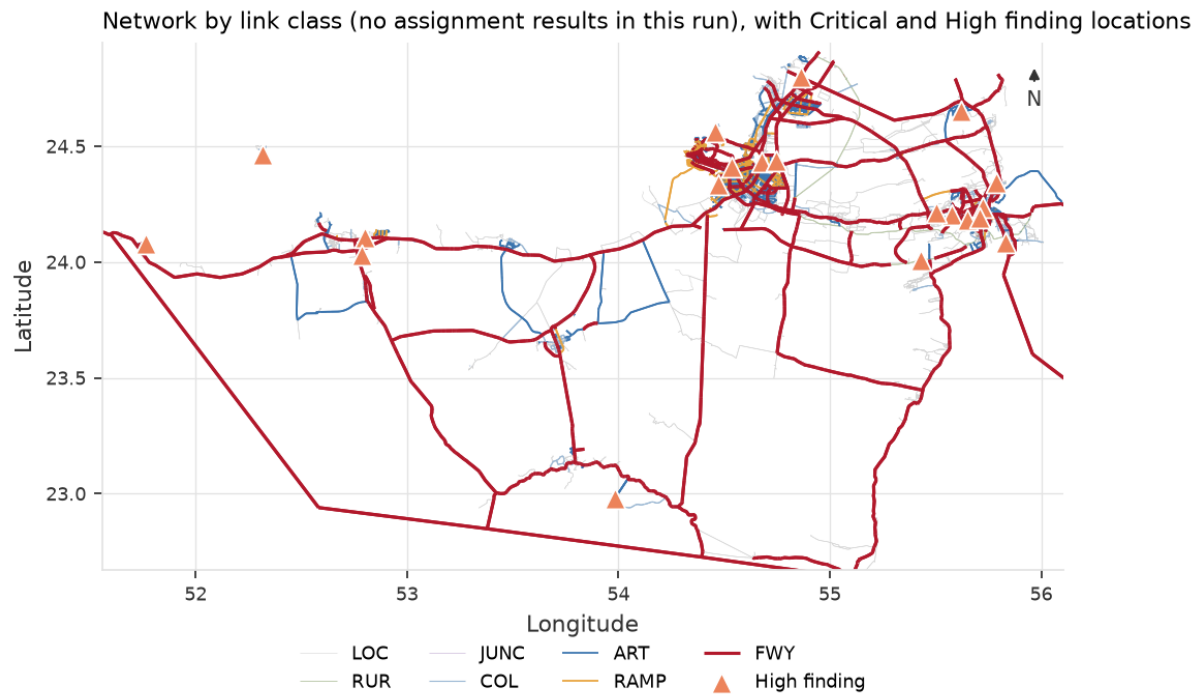
Key performance indicators

KPIs with their published definitions

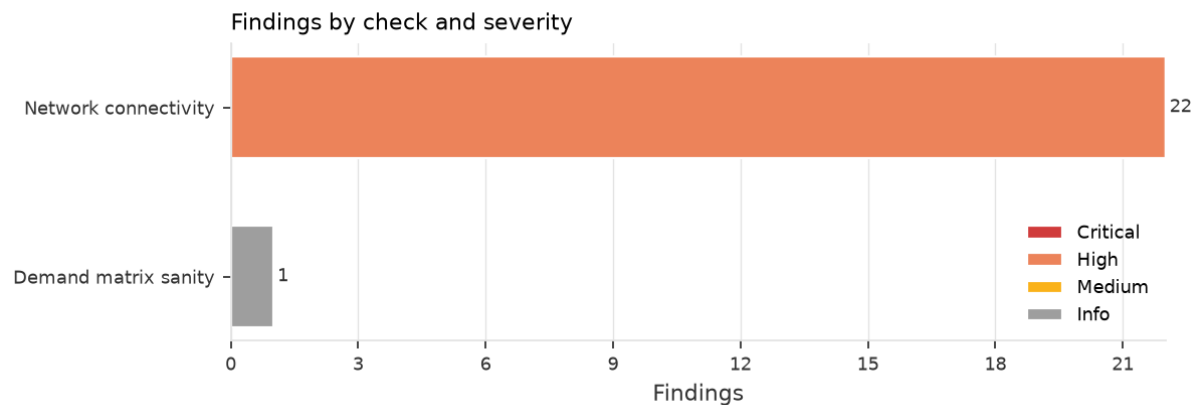
KPI	Value	Period	Definition
Link length, each direction	32221.1 km	all	Sum of link length over all highway links (each direction counted). SQL: SELECT SUM(length_m) / 1000.0 AS value FROM links
Lane-km, all roads	56,839 lane-km	all	Sum of link length x lanes over all highway links. SQL: SELECT SUM(length_m * lanes) / 1000.0 AS value FROM links
Lane-km, freeways	24,012 lane-km	all	Sum of link length x lanes over links of class FWY. SQL: SELECT SUM(length_m * lanes) / 1000.0 AS value FROM links WHERE link_class = 'FWY'
Total demand trips, all periods	12,731,560 trips	all	Sum of all DEMAND matrix cells, every purpose, mode and period. SQL: SELECT SUM(trips) AS value FROM od WHERE matrix_kind = 'DEMAND'
Population	5,934,196 people	all	Sum of the POP land-use variable over all zones. SQL: SELECT SUM(value) AS value FROM land_use WHERE variable = 'POP'

[src: kpis.json; config/kpis.yaml]

Maps and charts



Network map: links by class (the run has no assignment results), Critical and High finding locations marked
[src: links table (geometry_wkt, link_class); findings.json]



Findings by check and severity

[src: findings.json; 23 findings]

Check coverage

13 checks were run: 4 completed, 9 skipped because a required table was missing, 0 ended in error.

[src: check_results.json]

Check coverage

Check	Status	Findings	Message	Duration (s)
Null values and identifier integrity	ok	0		0.11
Network connectivity	ok	22	disconnected_component: 2,659 occurrences grouped into one summary plus the 10 most severe; node_id_mismatch: 94 occurrences grouped into one summary plus the 10 most severe	2.48
Centroid connector placement	ok	0		0.03
Transit line integrity	skipped	0	required table(s) missing: transit_lines, transit_segments	0.00

Check	Status	Findings	Message	Duration (s)
Land-use control totals	skipped	0	required table(s) missing: control_totals	0.00
Link volume outliers	skipped	0	required table(s) missing: link_flows	0.00
Convergence and noise band	skipped	0	required table(s) missing: convergence	0.00
Demand matrix sanity	ok	1		0.16
Trip length distribution	skipped	0	required table(s) missing: skims	0.00
Unrealistic speeds and times	skipped	0	required table(s) missing: link_flows	0.00
Parameter drift	skipped	0	required table(s) missing: parameters	0.00
Unexplained demand shift	skipped	0	base run lacks table(s): od	0.00
Unused transit services	skipped	0	required table(s) missing: line_loads, transit_lines, transit_segments	0.00

[src: check_results.json]

DRAFT

Appendix

Method notes

Each check maps a computed score to a severity band. Bands are checked top-down and the first match wins. The thresholds below are read from config/checks.yaml (defaults: enabled=True, max_findings_per_check=200, group_issue_above=25, group_keep_worst=10).

[src: config/checks.yaml]

Check thresholds and severity rules

Check	Enabled	Parameters	Severity rules
null_and_id_integrity	yes	required_columns = {'links': ['link_id', 'a_node', 'b_node', 'length_m', 'link_class', 'capacity_vph', 'ffs_kph'], 'nodes': ['node_id', 'x', 'y'], 'zones': ['zone_id'], 'link_flows': ['link_id', 'period', 'user_class', 'volume']}	Critical when missing_node_refs > 0; Critical when missing_zone_refs > 0; High when missing_link_refs > 0; High when null_share > 0.01; Medium when null_share > 0
network_connectivity	yes	min_capacity_vph = 1; min_ffs_kph = 5; touch_tolerance_m = 1; near_gap_m = 50	Critical when issue == 'zero_capacity' or issue == 'zero_speed'; High when issue == 'node_id_mismatch'; Medium when issue == 'orphan_node'; High when issue == 'oneway_dead_end'; High when issue == 'disconnected_component'
centroid_connectors	yes	high_class_links = FWY, EXP; connector_classes = CONN; max_connector_length_m = 5,000	High when issue == 'connector_on_high_class'; Medium when issue == 'connector_too_long'
transit_line_integrity	yes	headway_min_range = 2, 120; modes_headway_max = {'METRO': 20, 'LRT': 30, 'BRT': 30, 'BUS': 120, 'RAIL': 120, 'TRAM': 30, 'FERRY': 120}	Critical when issue == 'broken_sequence'; High when issue == 'headway_out_of_range'; Critical when issue == 'node_not_in_network'
land_use_control_totals	yes	tolerance_share = 0.02	Critical when abs_diff_share > 0.10; High when abs_diff_share > 0.05; Medium when abs_diff_share > 0.02
link_volume_outliers	yes	vc_critical = 1.3; vc_high = 1.1; vc_medium = 0.95; zscore_threshold = 3; min_vc_for_outlier = 0.25; min_group_size = 20; low_volume_share_of_group_median = 0.02; user_class = ALL; periods = AM, PM; exclude_link_classes = CONN	Critical when vc_ratio >= 1.30; High when vc_ratio >= 1.10; High when abs_zscore >= 4.0; Medium when vc_ratio >= 0.95 or abs_zscore >= 3.0; Medium when too_low
convergence_and_noise	yes	rel_gap_target = 0.001; rel_gap_high = 0.01; geh_stable = 1; geh_stable_share_target = 0.95; noise_band_iterations = 3; stages = HWY_ASSIGN, PT_ASSIGN	Critical when rel_gap > 0.01; High when rel_gap > 0.001; Medium when stable_share < 0.95
matrix_sanity	yes	intrazonal_share_max = 0.15; row_col_ratio_max = 5; min_zone_total_for_ratio = 50; pa_format_share = 0.25; fractional_cell_max_trips = 0.01; fractional_share_min = 0.01	Critical when issue == 'negative_cells'; High when issue == 'row_col_imbalance'; Medium when issue == 'intrazonal_share'; Info when issue == 'fractional_artefact'
trip_length_distribution	yes	max_mean_diff_share = 0.15; max_mean_diff_share_high = 0.3; distance_bins_km = 0, 2, 5, 10, 20, 40, 80, 200; skim_dist_units = km; reference_mean_km = {}	High when mean_diff_share > 0.30; Medium when mean_diff_share > 0.15
unrealistic_speeds_times	yes	min_cong_speed_kph = 5; max_speed_over_ffs_ratio = 1.05; max_delay_s = 900; periods = AM, PM; user_class = ALL	High when issue == 'speed_above_ffs'; High when issue == 'speed_below_floor'; Medium when issue == 'delay_outlier'
parameter_drift	yes	numeric_tolerance = 1e-09	High when drifted
unexplained_demand_and_shift	yes	sector_shift_share = 0.1; catchment_km = 5; min_sector_pair_trips = 100; explain_land_use_share = 0.01	High when shift_share > 0.25; Medium when shift_share > 0.10
unused_transit_services	yes	min_load_factor = 0.05; min_catchment_demand = 500; catchment_km = 1; pt_modes = PT	High when load_factor < 0.02; Medium when load_factor < 0.05

[src: config/checks.yaml]

Glossary

Glossary

Term	Meaning
V/C	Volume-to-capacity ratio: link volume per hour divided by link capacity per hour. Values above 1.0 mean demand exceeds capacity.
GEH	A statistic comparing two flows, M and C: $\sqrt{2(M - C)^2 / (M + C)}$. Values below 5 are usually treated as a good match.
Relative gap	Assignment convergence measure: the excess cost of the current route choices over the cheapest possible routes, divided by total cost. Smaller is better.
Noise band	The size of link flow change between the last assignment iterations. Differences inside the band are not treated as significant.
Centroid connector	An artificial link that loads a zone's trips onto the network at a chosen node.
Screenline	An imaginary line across the network on which modelled and observed crossings are compared.

[src: STEAM-AI glossary]

DRAFT